OUTCOMES-BASED EDUCATION (OBE) COURSE SYLLABUS

CSci 21
Data Structures and Algorithms

Quality Goals of the Faculty of Computing

1. Equip students with global competencies in computing by offering inclusive, high-quality education through updated, industry-relevant curriculum that meets international standards, supported by cutting-edge facilities and highly qualified, well-trained faculty and administrative staff.
2. Advance knowledge and technology in computing and related sciences through impactful research conducted in collaboration with national and international partners, focusing on sustainable development, climate change mitigation, food security, and technological innovation.
3. Contribute to societal transformation through community engagement by leveraging computing technology to address societal challenges, promote sustainable agriculture, strengthen science and technology, and drive national and global development.

Quality Objectives of the Department of Computer Science and Technology

1. To provide relevant undergraduate and graduate curricular programs in computer science and allied fields.
2. To develop a strong and dynamic faculty and staff in computer science and allied fields.
3. To produce highly employable graduates in computer science and allied fields.
4. To achieve long-term linkages and cooperation with local, national, international institutions and agencies involved in the development and utilization of information and communication technology.
5. To provide technical assistance in the utilization of information and communication technology.
6. To cooperate with other units within the university to support developmental programs.

I. PROGRAM INFORMATION

1. Name of the Program	Bachelor of Science in Computer Science
2. CHED CMO Reference	CHED CMO No. 25 s. 2015
3. BOR Approval	BOR Resolution No. 76 s. 2018

4. Program Educational Objectives and Relationship to Institution Mission

Program Educational Objectives	Mission*		
	a	b	c
1. Industry practitioners, entrepreneurs, and/or skilled workers with technical certifications related to information technology.	✓	✓	
2. Responsible leaders in IT professional organizations showcasing relevant technical accomplishments and recognized contributions.	✓	✓	✓
3. Researchers engaging in advanced research and studies in computing sciences providing people-centered, environment-friendly, and community-focused solutions.	✓	✓	✓

**a - produce graduates equipped with advanced knowledge and lifelong learning skills, b - ethical standards through high quality instruction and innovative research, c – impactful community engagements*

II. COURSE INFORMATION

1. Course Code	CSci 21
2. Course Title	Data Structures and Algorithms
3. Pre-requisite	CSci 14 Intermediate Programming
4. Co-requisite	None
5. Credit	3 units
6. Semester Offered	1st semester
7. Number of hours	2 hours lecture and 3 hours laboratory per week
8. Course Description	The course covers the standard data representation and algorithms to solve computing problems efficiently. This covers the following: arrays, linked lists, stacks, queues, hash tables, and trees.

12. Program Outcomes (POs) **in relation to the Program Educational Objectives** (PEOs)

Program Outcomes (POs)		**Program Educational Objectives**		
		1	2	3
1	Develop computing models by applying appropriate computing fundamentals, computing specializations, mathematics, science, and domain knowledge from defined problems and requirements.	✓		
2	Identify, analyze and solve simple and complex computing problems and requirements reaching substantiated conclusions through literature research and formulation.			✓
3	Create computer-based systems applying mathematical foundations, algorithmic principles, and computer science theory and demonstrating comprehension of the tradeoffs involved in design choices.	✓		
4	Analyze information security issues related to the design, development, and use of information systems using critical thinking skills to identify vulnerabilities and threats.	✓		
5	Design and evaluate effective solutions, systems, components, or processes that meet specified needs while incorporating appropriate consideration for public health and safety, cultural, societal, and environmental factors.			✓
6	Use modern computing tools effectively in conducting complex computing activities, while being aware of their limitations, to achieve a common goal.	✓		✓
7	Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary settings.		✓	
8	Communicate effectively with the computing community and with society at large about computing activities by comprehending and writing effective reports, design documentation, make effective presentations, and give and understand clear instructions.		✓	✓
9	Recognize the legal, social, ethical, and professional issues involved in the utilization of computer technology and be guided by the adoption of appropriate professional, ethical, and legal practices.		✓	
10	Engage in independent learning for continual development as a computing professional.			✓

13. Course Outcomes (COs) and Relationship to Program Outcomes (POs)										
After completing this course, the student must be able to perform the following COs:	*Program Outcomes Code*									
	1	2	3	4	5	6	7	8	9	10
CO1. Solve problems that utilizes array and linked list data structure	E	E	E							
CO2. Solve problems that utilizes stack and queue data structure	E	E	E							
CO3. Solve problems that utilizes hash table data structure	E	E	E							
CO4. Solve problems that utilizes tree data structure	E	E	E							

14. Course Content and Plan					
Week	**Topics**	**Learning Outcomes**	**Teaching and Learning Activities**		**Assessment Tasks**
			Teaching Activities	**Learning Activities**	
Class Orientation					
1	1. OBE Syllabus 2. Requirements 3. Assessment System and Activities 4. Class Policies 5. Safety Guidelines and Emergency Response during fire and earthquake Values: <u>Accountability, Integrity, Leadership, Trust, and Self-discipline</u>		• Setting of expectations • Agreements • Learning mode • Mode of communication • Familiarization of VSUEE/VC*		
CO1: Solve problems that utilizes array and linked list data structure					
2 - 5	Module No. 1 Arrays Module No. 2 Linked List	LO 1.1 Explain memory representation of arrays LO 1.2 Implement insertion, deletion, and search operations in arrays LO 1.3 Analyze time complexity in array operations LO 1.4 Diagnose array-related implementation errors LO 2.1 Explain memory representation of linked list LO 2.2 Implement insertion, deletion, and search operations in linked list LO 2.3 Analyze time complexity in linked list operations LO 2.4 Diagnose linked list-related implementation errors	Lecture Live coding Interactive discussion	Computing problem solving Code review with peers	Activity 1 Activity 2 Activity 3 Quiz 1 Practical Exam 1

	Module No. 3 Arrays vs. Linked List Values: Knowledge and Understanding	LO 3.1 Justify optimal structure choices using complexity analysis LO 3.2 Implement accurate solutions for problems using arrays and/or linked lists			
CO2: Solve problems that utilizes stack and queue data structure					
6 - 8	Module No. 4 Stacks Module No. 5 Queues Values: Knowledge and Understanding	LO 4.1 Implement accurate stack operations LO 4.2 Implement accurate solutions for problems using stack ADT LO 4.3 Diagnose stack-specific errors LO 5.1 Implement accurate queue operations LO 5.2 Implement accurate solutions for problems using queue ADT LO 5.3 Diagnose queue-specific errors	Lecture Live coding Interactive discussion	Computing problem solving Code review with peers	Activity 4 Activity 5 Quiz 2 Practical Exam 1
9	**MIDTERM EXAMINATION**				
CO3: Solve problems that utilizes hash table data structure					
10 - 12	Module No. 6 Hash Table Values: Knowledge and Understanding	LO 6.1 Implement accurate hash table operations LO 6.2 Implement accurate collision resolution techniques LO 6.3 Analyze time complexity under different load factors LO 6.4 Implement accurate solutions for problems using hash table ADT	Lecture Live coding Interactive discussion	Computing problem solving Code review with peers	Activity 6 Quiz 3 Practical Exam 2
CO4: Solve problems that utilizes tree data structure					
13 – 17	Module No. 7 Introduction to Trees Module No. 8 Binary Search Trees Values: Knowledge and Understanding	LO 7.1 Identify valid properties of a binary tree LO 7.2 Perform preorder, inorder, postorder traversals given binary tree definition LO 7.3 Synthesize a valid postfix expression from a binary expression tree LO 8.1 Implement accurate binary search tree operations LO 8.2 Compare and contrast BST to the generic binary tree	Lecture Live coding Interactive discussion	Computing problem solving Code review with peers	Activity 7 Activity 8 Quiz 4 Practical Exam 2
18	**FINAL EXAMINATION**				
** VSUEE/VC – VSU E-Learning Environment/ Virtual Classroom*					

15. Life-long Learning Opportunities

1. Foundational Skill Reinforcement
   a. Algorithmic thinking is useful especially in competitive programming
   b. Tradeoff analysis is significant in architectural decision-making in back-end systems
   c. Memory management promotes continuous engagement with systems programming (OS kernels and embedded systems)
2. Cross-Disciplinary Adaptation
   a. Suffix trees for genome sequencing in Bioinformatics
   b. Merkle trees in Blockchain
   c. Machine learning architectures
3. Teaching and Mentorship

16. Contribution of Course to Meeting the Professional Component (%)

Professional Education*: 2.75%*

17. References and Other Learning Resources

A. Textbook(s)

1. Barnett, G., & del Tongo, L. (2008). *Data Structures and Algorithms: Annotated Reference with Examples* (1st ed.).
2. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). *Introduction to Algorithms* (4th ed.). MIT Press.
3. Gilberg, R. F., & Forouzan, B. A. (2005). *Data Structures: A Pseudocode Approach with C*. Thomson Course Technology.
4. Harris, S., & Ross, J. (2006). *Beginning Algorithms*. Wiley.
5. Kernighan, B. W., & Ritchie, D. M. (1988). *The C Programming Language* (2nd ed.). Prentice Hall.
6. Sedgewick, R. (1998). *Algorithms in C: Parts 1–4* (3rd ed.). Addison-Wesley.
7. Skiena, S. S. (2008). *The Algorithm Design Manual* (2nd ed.). Springer.
8. Wengrow, J. (2017). *A Common-Sense Guide to Data Structures and Algorithms*. Pragmatic Bookshelf.

B. Videos

1. MyCodeSchool (Best for visual explanations & C implementations)
   https://youtube.com/playlist?list=PL2_aWCzGMAwI3W_JlcBbtYTwiQSsOTa6P
2. Understanding and implementing a Linked List in C and Java
   https://www.youtube.com/watch?v=VOpjAHCee7c
3. Priority Queue Introduction
   https://www.youtube.com/watch?v=wptevk0bshY
4. How to Create a Hash Table Project C++
   https://www.youtube.com/watch?v=MfhjkfocRR0&list=PLTxllHdfUq4f7-uHOpxXnBUbsuLiI9pmb
5. 5.10 Binary Search Trees (BST) - Insertion and Deletion | DSA Full Course
   https://www.youtube.com/watch?v=cySVml6e_Fc
6. Using Leaks (the valgrind equivalent) on Mac (Shown on M1) to detect memory leaks
   https://www.youtube.com/watch?v=bhhDRm926qA
7. Introduction to Big O Notation and Time Complexity
   https://www.youtube.com/watch?v=D6xkbGLQesk

F. Problem Solving Sites

1. LeetCode (https://leetcode.com/)
2. HackerRank (https://www.hackerrank.com/)
3. Codeforces (https://codeforces.com/)

18. Course Assessment and Evaluation

The performance of students will be assessed and evaluated based on the following:

Item No,	Assessment Tasks	Percentage Contribution (1)	No. of Times in the Semester (2)	Individual Task % Contribution (1/2)
1	Activity	20	8	2.5%/A
2	Quiz	20	4	5%/Q
3	Practical Examination	30	2	15%/PE
4	Term Examination	30	2	15%/TE
		100%		

Grading System (60% Passing)

Range	*Grade*	*Range*	*Grade*
100.00 – 96.00	1.00	77.99 – 73.50	2.25
95.99 – 91.50	1.25	73.49 – 69.00	2.50
91.49 – 87.00	1.50	68.99 – 64.50	2.75
86.99 – 82.50	1.75	64.49 – 60.00	3.00
82.49 – 78.00	2.00	Below 60.00	5.00

19. Course Policies

- **VSU E-Learning Portal** (https://elearning.vsu.edu.ph/) is the official learning portal for this course.
- This syllabus is designed for face-to-face lecture class.
- Deadlines are absolute unless extended due to any force majeure e.g., natural disasters, hospitalization (inpatient), sickness (outpatient), war, suspension of classes via VSU OP memoranda. In the event of sickness (outpatient), it must be supported with a medical certificate signed by the doctor with his license number attached, otherwise the claim will not be accepted.
- University policies are strictly followed. Please be guided accordingly.
- Lastly, as we embark in this “new normal”. Let us have an open mind and heart as we adjust in this new way of delivering the teaching-learning process and continue to aim for the highest quality in computer science education.

This class policy serves as our written agreement for the whole semester/midyear. If there are any changes to enhance the class learning opportunity within the semester/midyear, it will be communicated accordingly.

20. Course Materials and Facilities Available

Faculty:	Student:
• Whiteboard, marker, and eraser • Laptop • C compiler (GCC) • Projector	• VSUEE (E-learning portal) • Safe Exam Browser (SEB) • Electronic reading materials • PC • C compiler (GCC)

21. Revision History				
Revision number	**Date of Revision**	**Date of Implementation**	**Highlights of Revision**	**Revised by**
00	Dec. 28, 2018	Aug. 1, 2019	New syllabus creation	Jomari Joseph Barrera
01	Oct. 2, 2019	Jan. 2, 2020	ISO OBE syllabus format transfer	Jomari Joseph Barrera
02	May 6, 2020	Aug. 10, 2020	Revised for the blended mode of learning (pandemic season)	Jomari Joseph Barrera
03	Aug. 24, 2022	Sep. 12, 2022	Revision of topics, assessment tasks, learning mode and grading system	Jomari Joseph Barrera
04	Mar. 21, 2024	Aug. 12, 2024	Revision of refences and learning outcomes	Jomari Joseph Barrera
05	May 30, 2025	June 13, 2025	Topics harmonized under Faculty of Computing Update of references and other learning resources Revision of assessment system	Jomari Joseph Barrera

22. Preparation			
	Name	Signature	Date Signed
Prepared by	Jomari Joseph A. Barrera		

III. INSTRUCTOR/PROFESSOR INFORMATION

1. Name of Instructor/Professor	Jomari Joseph A. Barrera
2. Office and Department	Department of Computer Science and Technology
3. Telephone/Mobile Numbers	(053) 565 0600 loc. 1022 / +63 970 068 4506
4. Email Address	jomarijoseph.barrera@vsu.edu.ph
5. Consultation Time	Wednesday 10 – 12, 1 – 3 PM

6. Department Instructional Materials Review Committee:

Committee	Name	Signature	Date Signed
Member:	**JUDE B. ROLA**		
Member:	**MAGDALENE C. UNAJAN**		
Chairperson:	**JONAH FLOR O. MAAGHOP**		

	Name	Signature	Date Signed
Noted by:	**MAGDALENE C. UNAJAN** Head, DCST		
Verified by:	**JUDE B. ROLA** Dean, Faculty of Computing		
Validated by:	**MARK GIL A. VEGA** Head, IMDO		

Note:

1) The number of POs will depend on each degree program offered
2) COs and Relationship to POs
    a. (I) - **Introductory** – an Introductory Course to an outcome
    b. (E) - **Enabling** – an Enabling Course or a course that strengthens the outcome
    c. (D) - **Demonstrated** – a Demonstrative Course or a course demonstrating an outcome.

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(2) Distribution of copies: Faculty, IMDO